Gear Lab

1) What is the purpose of a transmission?

   A transmission's job is to optimize the power source such as a motor to a load. It makes the power into both translational speed and torque. A transmission can also change the direction of the load.

2) Is this a compound or simple gear train?

   This gear train is a compound gear train because it has multiple gears included, specifically it has 2 bevel gears and one big and one small gear.

3) What is the overall speed ratio $\frac{\omega_{out}}{\omega_{in}}$?

   It is 96/11 for the overall speed ratio, this is because the motor is connected to a bevel gear with 48 teeth and then it is next to one with 24 teeth so this pair has speed ratio of 2 (N/N= 48/24), and sine the second bevel gear is on the same axle/shaft, this means that the big gear has the same angular velocity as the bevel gear(24 teeth). So to get the overall speed ratio you can just multiply the speed ratio of 48/24 to the 96/22, which becomes 96/11

4) What is the overall torque ratio $\frac{\tau_{out}}{\tau_{in}}$?

   Torque ratio is the inverse of the speed ratio so it is 11/96.

5) What is the overall Power ratio $\frac{P_{out}}{P_{in}}$?

   If there is no heat loss(i.e. the system is ideal), then the power ratio is equal to 1, but if we account for friction or heat generated elsewhere, the power ratio is less than 1(but only a little bit in comparison to belt drive).

6) What is the purpose of the locating pins?

   Locating pins help with alignment but whereas, screws might have threads that make the alignment a bit inaccurate causing the gears to not line up, the locating pins ensure that the gears in the gear lab are in the exact position they need to be in.

7) For the Diamond pin and its Bushing, Look up the parts on McMaster. Compare the Tolerance ranges of the Bushing Inner Diameter and the Pin Outer Diameter. What “fit class” is used for this Pin and Bushing pair?

   I think it is the Runnig fit because it is pretty common and looks like it allows the pin to be inserted.

   The diamond pins outer Diameter is 0.3125 in with a tolerance that ranges -0.0008 to -0.0005. The ommer diameter is 0.0013 which lets the pin move a little.

8) For the Bushing Outer Diameter, what “fit class” would be appropriate for a press fit inside the aluminum plate?

   The fit class is in FIN2, which is force fit class 2 and it would be 0.0005 to 0.0013 as that falls within the clearance needed to hold it.

9) For #8 above, what should the hole dimension with tolerance be?

   If we have a Force fit Class 2 then the hole dimension we need would is -0.007, which would be 0 to 0.009 allowing some interference and a steady hold.

10) What type of bearing is used?

    It is a ball bearing

11) What is the purpose of the springs between the bearings?

    Allowed tolerance, locate the bearing on one side, and torsion and bearing preload

12) What is the size of the Socket Head Cap screw used in the assembly? Check with calipers and thread gage.

    measuring it it has a major diameter of .26(1/4”) inches and 1 inch length. This means it is a #10-24 UNC x 1” socket head cap screw as it has 20 threads per inch and

13) What would be the size of the tap drill used to drill the hole for the screw in #12? Verify with calipers. .

    .201 in with #7 drill bit

14) What is the size of the clearance drill used for the screw in #12? Verify with calipers.

    0.257 in. to 0.281 in.

15) What did you learn or find interesting about this lab?

    This lab is a bit difficult to view working together as a gear. Like it is clear once it is built but understanding what goes where and getting everything located corrected was a pain but once I got it, it was very easy to see which is cool.

16) What would you like to know more about?

<u>The importance of gears and how to know where to put pins to have the gear aligned tgether</u>

Belt Lab

1) What is a general advantage of a belt/chain drive compared to a gear drive? This should come from your observations, not necessarily from the lecture materials.

   Low maintenance and the belt can operate a bit more efficiently/smoother in comparison to the gear one.

2) What is one specific advantage of a belt drive?

It can transmit power over long distances,

3) What is one specific disadvantage of a belt drive?

The belt drive is not the best for high power applications because belts can slip off or wear off from heat.

4) What is one specific advantage of a gear drive?

Gears can allow for an ideal speed ration since it is not slipping like a belt drive could.

5) What is one specific disadvantage of a gear drive?

A gear drive needs precise alignment like with locating pins because misalighnment is less effective and can wear the gears.

6) What is the overall speed ratio $\frac{\omega_{out}}{\omega_{in}}$?

-1/3 * -3/4 = 0.208

7) What is the overall torque ratio $\frac{\tau_{out}}{\tau_{in}}$?

1/0.208 = 4.808

8) What is the overall Power ratio $\frac{P_{out}}{P_{in}}$?

Power should be 1 if no energy is lost, but since belts are more susceptible to slippage there is most likely a power ratio less than 1.

9) What is the purpose of the spring-loaded tensioning arms?

Provide tension on the belt automatically

10) What is the size of the Socket Head Cap screw used in the assembly? Check with calipers and thread gage.

It was an M8 x 1.25 socket head cap screw and has a 13mm head diameter with 8mm thread size and a corse pitch of 1.25mm

11) What would be the size of the tap drill used to drill the hole for the screw in #10? Verify with calipers. .

    To work with M8x 1.25 socket head cap screw should be 6.8mm tape drill.

12) What is the size of the clearance drill used for the screw in #10? Verify with calipers.

    8.8mm

13) What types of springs are used?

    A compression spring and tension springs.

14) What did you learn or find interesting about this lab?

    I learned that all components allow for the whole system to work and without one like the springs then the belt wouldn't work at all.

15) What would you like to know more about?

    How to set up belt drive system. Like so I can learn how to do it on my own and design a setup like perhaps for the car project.